

Exploring star formation in SDSS galaxies

A companion to my 2023 Light My MOSFIRE internship notebook

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I wrote this guide to give some context to `IAstro 2023-Light My MOSFIRE.ipynb`: why I made the choices I did, and how the tasks fit together. The notebook contains the calculations, plots and working code, so I will leave most of the line-by-line detail there.

The spectra used in this notebook are of low-redshift galaxies from the [Sloan Digital Sky Survey \(SDSS\)](#), not the MOSFIRE instrument. The project's broad question was whether galaxies in different parts of the star-forming main sequence show differences in their fitted properties. We got as far as building the sample, making the diagnostic plots and preparing spectra for FADO.

A note about the files: The starting FITS tables were supplied during the internship, and I cannot redistribute them. This repository is a record of my analysis, but can be used to reproduce similar results with similar starting data and FADO.

The question behind the notebook

A galaxy's [star formation rate](#) (SFR) describes how quickly it is forming stars now. Its [stellar mass](#) tells a different, longer-term story: how much mass it has built up in stars. When we plot SFR against stellar mass, many star-forming galaxies occupy a band called the *star-forming main sequence*. Galaxies above and below that band are interesting, but their position alone does not tell us why they are there. The relation also depends on the population and redshift being studied [4].

That is where spectra become useful. They carry information about the ionized gas and the mixture of stellar populations contributing to a galaxy's light. We wanted to select galaxies from different parts of the SFR–mass plane, examine their spectra with FADO, and see whether properties such as age, metallicity and nebular contribution changed across those selections [1].

FADO and where to find it

FADO stands for *Fitting Analysis using Differential evolution Optimization*. It was developed by Jean Michel Gomes and Polychronis Papaderos at the [Instituto de Astrofísica e Ciências do Espaço](#) in Portugal. Its approach fits the stellar and nebular contributions to galaxy spectra self-consistently. If you use FADO in research, the creators ask that you cite their presentation paper, Gomes and Papaderos (2017) [1].

The notebook uses the original FADO executable and its FITS outputs. The [IA FADO page](#) links the [v1b package download](#), the [v1b manual](#), and the [distribution archive](#). The package contains its own ReadMe and Read_F instructions, configuration files, stellar-population library and test spectra. The official page describes v1b as its latest stable distribution, released in 2018 [2].

There is also a [PyFADO beta listing on PyPI](#). Its description says it includes Python tools for reading and plotting FADO outputs. I have linked it for anyone interested in exploring it, but the repository and quick-start links on that listing were unavailable when I checked them. This notebook does not use PyFADO, and the PyPI listing should not be treated as a drop-in replacement for the FADO executable [3].

Before Task 01: bringing the measurements together

The first step is practical: the emission lines, redshifts, stellar masses and other measurements need to refer to the same galaxies. The notebook brings together the supplied `FADO_SDSS.fits`, `FADO_Emission.fits`, `SDSS_Params.fits` and `FADO_Conversion.fits` tables, using the index in the conversion table. It saves the combined working table as `FADO_ALL.fits`.

The alignment of those rows matters more than the mechanics of joining FITS files. If the rows do not match, a plot can still look reasonable while combining measurements from different objects.

Task 01: choosing a usable sample

The next steps rely on the $H\alpha$ and $H\beta$ [emission lines](#). Young, massive stars can ionize nearby gas; when hydrogen recombines, it produces lines that include $H\alpha$ and $H\beta$. Their relative strengths will later help with a dust correction, but the comparison is not useful if either line is too noisy.

The notebook calculates the signal-to-noise ratio for each line as

$$(S/N)_{H\alpha} = \frac{F_{H\alpha}}{\sigma_{F_{H\alpha}}}, \quad (S/N)_{H\beta} = \frac{F_{H\beta}}{\sigma_{F_{H\beta}}}. \quad (1)$$

We therefore kept galaxies with a signal-to-noise ratio greater than 3 in both lines and applied the notebook's stellar-mass cut of $\log M_{\text{cbst}} > 5$. This reads `FADO_ALL.fits` and saves `selected_galaxies.fits`. It also saves `FADO_ALL_SNR.fits`, which retains the full working table with the calculated signal-to-noise columns. The cut on stellar mass is a choice for this sample, not a general definition of a dwarf galaxy.

Task 02: identifying the star-forming galaxies

Before estimating an $H\alpha$ -based SFR, we needed to identify galaxies whose emission-line ratios are consistent with star formation. The notebook uses a [BPT diagram](#), plotting

$$x = \log_{10} \left(\frac{F_{[\text{N II}] 6584}}{F_{H\alpha}} \right), \quad y = \log_{10} \left(\frac{F_{[\text{O III}] 5007}}{F_{H\beta}} \right). \quad (2)$$

The two boundaries drawn in the notebook are

$$y_{\text{Kauffmann}} = \frac{0.61}{x - 0.05} + 1.30, \quad y_{\text{Kewley}} = \frac{0.61}{x - 0.47} + 1.19. \quad (3)$$

These come from Kauffmann et al. (2003) [7] and Kewley et al. (2001) [6], respectively. The original emission-line diagnostic goes back to Baldwin, Phillips and Terlevich (1981) [5].

Starting from `selected_galaxies.fits`, this task saves `sf_galaxies.fits`, `composite_galaxies.fits` and `agn_galaxies.fits`, along with `BPT_Diagram.png`. These names describe regions of this particular diagnostic plot. In particular, “AGN” here is the label used for objects above the Kewley curve; the notebook does not separately classify Seyferts and LINERs.

Task 03: from $H\alpha$ to the star-forming main sequence

For the star-forming sample, we used $H\alpha$ to estimate SFR. To correct for internal dust attenuation, the notebook uses the observed Balmer ratio, an assumed intrinsic ratio of 2.86, and the power-law shape associated with Charlot and Fall (2000) [8]:

$$k(\lambda) = \left(\frac{\lambda}{5500 \text{ \AA}} \right)^{-0.7}, \quad E_{\text{notebook}} = \frac{2.5}{k(4861 \text{ \AA}) - k(6563 \text{ \AA})} \log_{10} \left(\frac{F_{H\alpha}/F_{H\beta}}{2.86} \right), \quad (4)$$

$$A_{H\alpha} = E_{\text{notebook}} k(6563 \text{ \AA}), \quad F_{H\alpha, \text{corr}} = (F_{H\alpha} \text{ galsnorm } 10^{-17}) 10^{0.4 A_{H\alpha}}. \quad (5)$$

Here E_{notebook} labels the intermediate factor in the code; the dimensionless curve k is normalized at 5500 Å. The 10^{-17} factor puts the normalized line flux on the scale used by the notebook. For the assumed flat Λ CDM cosmology ($H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3$), Astropy obtains the luminosity distance $d_L(z)$. The last steps are

$$L_{H\alpha} = 4\pi d_L^2 F_{H\alpha, \text{corr}}, \quad \log_{10} \left(\frac{\text{SFR}_{\text{fib}}}{M_{\odot} \text{ yr}^{-1}} \right) = \log_{10} \left(\frac{L_{H\alpha}}{10^{41.31} \text{ erg s}^{-1}} \right). \quad (6)$$

The value $10^{41.31}$ is the calibration written in the notebook; absolute SFRs depend on the adopted stellar-population assumptions [9].

The `fibre_sfms()` function reads `sf_galaxies.fits`, saves a new FITS file under the name supplied to the function, and makes `SFMS_fib.png`. These are *fibre* estimates: the SDSS spectrum samples light through a limited aperture, rather than all of a galaxy’s light [10].

The next part adds the supplied logarithmic conversion columns to estimate *total* quantities:

$$\log_{10} \left(\frac{\text{SFR}_{\text{tot}}}{M_{\odot} \text{ yr}^{-1}} \right) = \text{LSFR} + \text{SFR_Conv}, \quad \log_{10} \left(\frac{M_{*, \text{tot}}}{M_{\odot}} \right) = \text{logMCbst} + \text{LSM_Conv}. \quad (7)$$

Both right-hand columns are already on logarithmic scales in this dataset. It saves a shortlisted FITS table and the plots `SFMS_total.png` and `SFMS_total_ver.png`. The straight lines drawn across the SFR–mass plane are working boundaries for choosing samples.

Task 04: choosing galaxies to examine more closely

We used two ways of dividing the star-forming sample: by position relative to the working main-sequence lines, and by stellar-mass range. The idea was to select galaxies from each region, fit their spectra, and compare their properties. The working SFR boundaries, using $m = \log_{10}(M_{*, \text{tot}}/M_{\odot})$ and $s = \log_{10}(\text{SFR}_{\text{tot}}/(M_{\odot} \text{ yr}^{-1}))$, are

$$s_{\text{upper}} = 0.77m - 6.70, \quad s_{\text{middle}} = 0.77m - 7.43, \quad s_{\text{lower}} = 0.77m - 8.20. \quad (8)$$

These are the lines coded for sample selection, rather than a universal fit to the main sequence. The four SFR regions and four mass ranges overlap; they are two ways of looking at the sample, not eight non-overlapping cells.

The `section()` function reads the shortlisted star-forming FITS table. For a chosen region it writes a text file with the sampled galaxies' identifiers and measurements, plus an IPAC-format coordinate file for the dust-map lookup. The intended comparison for this project was 50 galaxies per region, but the notebook does not contain a completed eight-region result or files, as only two regions were assigned per person. However, this notebook can be used to sample as many regions as needed.

Task 05: preparing spectra for FADO

There are two dust corrections in this workflow, and they have different jobs. Task 03 estimates *internal attenuation in the target galaxy* for the H α SFR. Here we prepare the full observed spectrum by correcting for *foreground Milky Way dust* along our line of sight. The notebook uses a colour-excess value from the [IRSA dust-map service](#) and the optical extinction law of Cardelli, Clayton and Mathis (1989) [11].

The code implements the optical Cardelli–Clayton–Mathis prescription [11] with $R_V = 3.1$ and $x = (\lambda/10^4 \text{ \AA})^{-1}$ in μm^{-1} , $y = x - 1.82$. For its optical range, it evaluates

$$a(x) = 1 + 0.17699y - 0.50447y^2 - 0.02427y^3 + 0.72085y^4 + 0.01979y^5 - 0.77530y^6 + 0.32999y^7, \quad (9)$$

$$b(x) = 1.41338y + 2.28305y^2 + 1.07233y^3 - 5.38434y^4 - 0.62251y^5 + 5.30260y^6 - 2.09002y^7, \quad (10)$$

$$\frac{A_\lambda}{A_V} = a(x) + \frac{b(x)}{R_V}, \quad f_{\lambda, \text{MW corr}} = f_{\lambda, \text{obs}} 10^{0.4E(B-V)R_V[a(x)+b(x)/R_V]}. \quad (11)$$

The same multiplicative factor is applied to the flux-density uncertainty. The observed wavelengths are also shifted by the galaxy's redshift. For a spectrum expressed as flux density per unit wavelength, the notebook changes the wavelength and flux-density arrays to its rest-frame representation:

$$\lambda_{\text{rest}} = \frac{\lambda_{\text{obs}}}{1+z}, \quad f_{\lambda, \text{rest}} = (1+z)f_{\lambda, \text{MW corr}}, \quad \sigma_{f, \lambda, \text{rest}} = (1+z)\sigma_{f, \lambda, \text{MW corr}}. \quad (12)$$

This step expresses the observed spectrum on a rest-frame wavelength axis; it does not by itself turn flux density into luminosity density [12].

`read_dust()` combines the dust-map values with the selected-galaxy list. The following functions retrieve the SDSS spectra and write one corrected, three-column text spectrum per galaxy, named `rest_spec_plate-mjd-fibre.txt`. They also make a plot of each spectrum before and after correction and write `gal_list_runname.txt`, which lists the spectrum filenames and luminosity distances needed for the FADO run.

Task 06: looking at the FADO results

FADO fits contributions from stars and ionized gas to a galaxy spectrum [1]. That makes it useful here: the earlier tasks tell us where a galaxy lies on the SFR–mass plane, while the fit offers more information about the populations and nebular emission behind its light. Light-weighted and mass-weighted ages, for example, answer different questions because young bright stars need not account for most of the stellar mass.

This part reads the per-galaxy FADO stellar and spectral output files (`.FADO_ST.fits` and `.FADO_1D.fits`). The notebook calculates a per-galaxy nebular percentage from the fitted spectral arrays:

$$f_{\text{neb}}(\%) = 100 \text{ median}_{\lambda: f_{\text{total}}(\lambda) > 0} \left[\frac{f_{\text{nebular}}(\lambda)}{f_{\text{total}}(\lambda)} \right]. \quad (13)$$

This is the median of the wavelength-by-wavelength fractions over pixels with positive total model flux. It then writes a CSV of measurements and a text summary of medians. An extra function plots the input spectrum alongside the total, stellar and nebular models.

References

- [1] Gomes, J. M. & Papaderos, P. (2017), *A&A* 603, A63. <https://doi.org/10.1051/0004-6361/201628986>.
- [2] Instituto de Astrofísica e Ciências do Espaço, *FADO tool and downloads*. <https://www.iaastro.pt/research/tools/fado.html>.
- [3] PyFADO beta listing, PyPI. <https://pypi.org/project/PyFADO/>.
- [4] Speagle et al. (2014), *ApJS* 214, 15. <https://arxiv.org/abs/1405.2041>.
- [5] Baldwin, Phillips & Terlevich (1981), *PASP* 93, 5. <https://doi.org/10.1086/130766>.
- [6] Kewley et al. (2001), *ApJ* 556, 121. <https://doi.org/10.1086/321545>.
- [7] Kauffmann et al. (2003), *MNRAS* 346, 1055. <https://arxiv.org/abs/astro-ph/0304239>.
- [8] Charlot & Fall (2000), *ApJ* 539, 718. <https://arxiv.org/abs/astro-ph/0003128>.
- [9] Kennicutt & Evans (2012), *ARA&A* 50, 531. <https://doi.org/10.1146/annurev-astro-081811-125610>.
- [10] SDSS, *Understanding SDSS spectroscopic data*. https://www.sdss3.org/dr9/spectro/spectro_basics.php.
- [11] Cardelli, Clayton & Mathis (1989), *ApJ* 345, 245. <https://ui.adsabs.harvard.edu/abs/1989ApJ...345..245C/abstract>.
- [12] Hogg et al. (2002), *The K correction*. <https://arxiv.org/abs/astro-ph/0210394>.